

1, a) Data mining is a process of extracting useful patterns and knowledge from large datasets

- Classification and clustering are two basic data mining tasks

KDD steps: Data Integration $\rightarrow$ Data Selection $\rightarrow$ Data Transformation

$\rightarrow$ Data Mining $\rightarrow$ Pattern evaluation $\rightarrow$ knowledge presentation

Role of preprocessing: Improves data quality, removes noise and missing values, increasing mining accuracy

b) Data cleaning: Process of correcting incomplete, inconsistent and inaccurate data

Two issues: Missing values, noisy data

Missing values: Replace with mean/median/mode or remove the record

Noisy data: Use binning, regression or clustering

2, a) Data Integration: Combining data from multiple sources into one consistent dataset

Challenges: Redundancy and Inconsistent

Redundancy: Remove duplicate records using correlation analysis

Inconsistency: Standardize formats and resolve naming conflicts

b) Covariance Analysis: Measures how two variables change together

Population Covariance:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3,

a,

- Both stock prices increase together
- Therefore, the covariance is positive, indicating they are positively correlated, not independent

b) Architecture of typical data mining system

Components:

- Database / Datawarehouse
- Data cleaning & Integration
- Data mining Engine
- Pattern Evaluation
- Knowledge base
- User Interface

These components work together to discover useful knowledge from data

4,

a, Find covariance:

$$x = \{4, 5, 6, 7, 9\}$$

$$y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean of } x = 6.2$$

$$\text{Mean of } y = 5.8$$

$$\text{Sample Covariance} = \text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$= -0.75$$

b, Major steps:

- Data Cleaning: detecting and correcting missing, incorrect or inconsistent data
- Data Integration: combining data from multiple sources into a single consistent data
- Data transformation: converting data into a suitable format for analysis or mining
- Data reduction: reducing the size of data while preserving its important information
- Data Discretization: converting continuous data into a finite number of intervals

Pipe line: Cleaning $\rightarrow$ Integration $\rightarrow$ transformation $\rightarrow$ reduction $\rightarrow$ Discretization

5, a,
Perform equal frequency binning

Dataset:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

Divide into 3 equal-frequency bins

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

b) Data discretization: converts continuous values into intervals

Importance: reduces data size, removes noise, improves mining efficiency

Equal-width

- Equal interval size
- Simple to implement
- May produce uneven bins

Equal-Frequency

- Equal number of records
- Handles skewed data better
- Balanced distribution of data